

DOCUMENT FOR THE OVERALL VOLTAGE MULTIPLIER's RESEARCH AND SIMULATION

So, for the harvester circuit the voltage multiplier plays a very important role, by which it takes the Alternating current which was harvested from the ambient rf signals present in the surrounding environment; Then it does two things:

- 1. It rectifies the input AC voltage into a DC voltage.**
- 2. It multiplies the input voltage level of the very weak rf signals which consist of 50 mV into almost depending upon how many stages we use of those multiplier circuit we can multiply it till a usable voltage level as above 1.5 V.**

Our aim is to maintain the output voltage of our harvester circuit to 1.8 V for a period of time sufficient to backscatter particular amount of data.

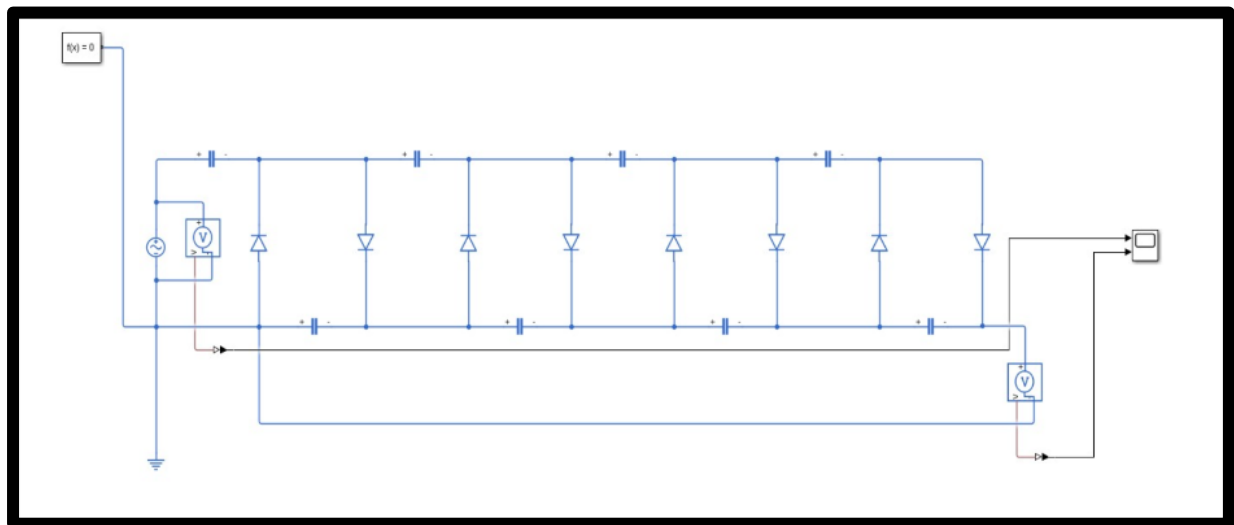
So, we initially got the hint of the voltage multiplier circuits by reading the 2013 paper where it was mentioned in their reference research paper. Which will multiply and rectify our input AC voltage into a usable voltage level and to make rf harvester self-dependable to power our microcontroller.

So we are going to see three different designs of harvester voltage multiplier circuits in this document and also we will see how we got to each multiplier circuit and how we simulated those circuit by using practical values.

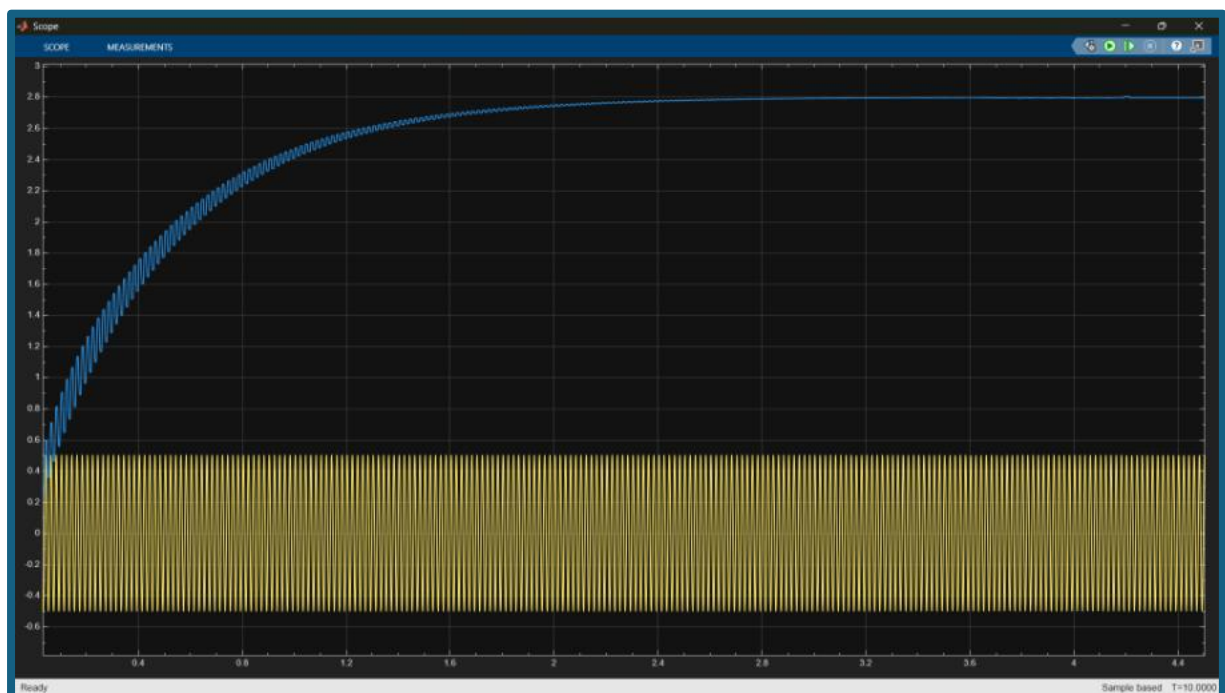
COCKCROFT-WALTON VOLTAGE MULTIPLIER

So, the first voltage multiplier circuit we got was Cockcroft-Walton Voltage Multiplier.

Here the circuit of a half wave Cockcroft-Walton voltage multiplier is:



And it's output voltage waveform:



Cockcroft-Walton voltage multiplier just multiplies the input AC voltage to steady DC high output voltage. So the following is the 4-stage Cockcroft-Walton voltage multiplier which multiplies the input low voltage for example 1V AC to output 6V DC, and if we increased the stage it will increase more till a point.

The formula of it is: $V_{out} = 2 \cdot n \cdot V_{peak} - 2 \cdot n \cdot V_f$

Where:

V_f = forward voltage of the diode

n = no. of stages

So in the simulation the parameters are taken based on practical real world application :

For capacitor:

1. Capacitor value = 100 uF
2. Series Resistance = 0.05 ohm

For Diode(low voltage Schottkey diode characteristics are considered):

1. $V_f = 0.25$ V
2. Resistance = 0.1 ohm So, by formula for example if we take 1 V

The output will be of 4 stage:

$$V_{out} = 2 \cdot 4 \cdot 1 - 2 \cdot 4 \cdot 0.25$$

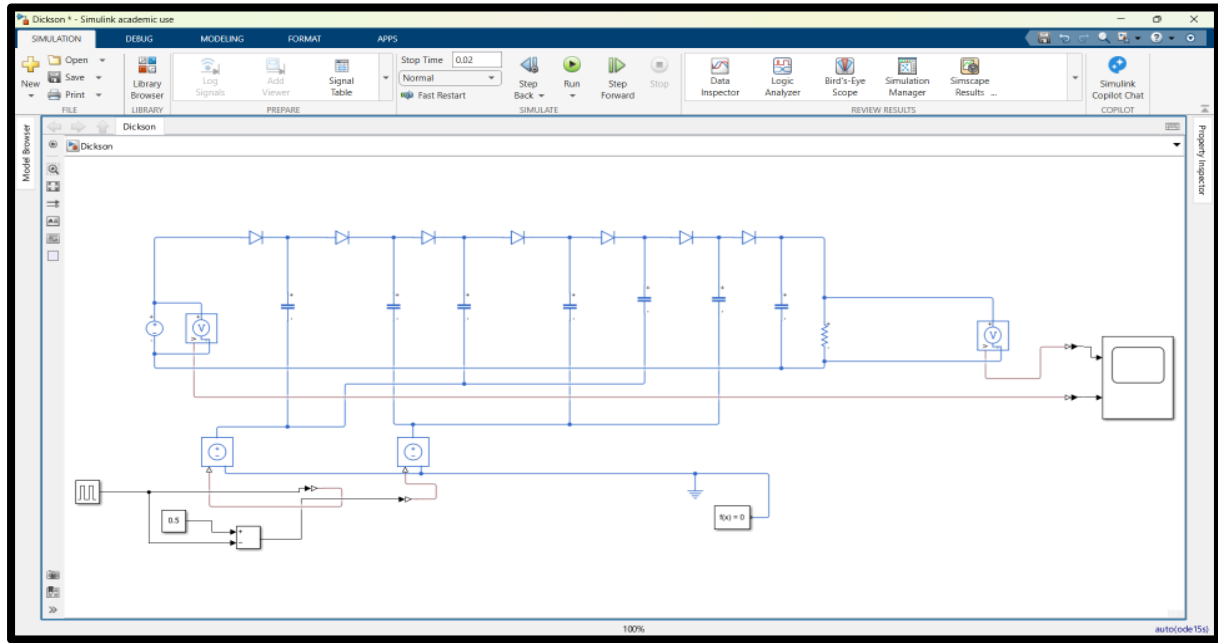
$$V_{out} = 8 - 2$$

$V_{out} = 6$ V, for 0.5 V it will be 2 V

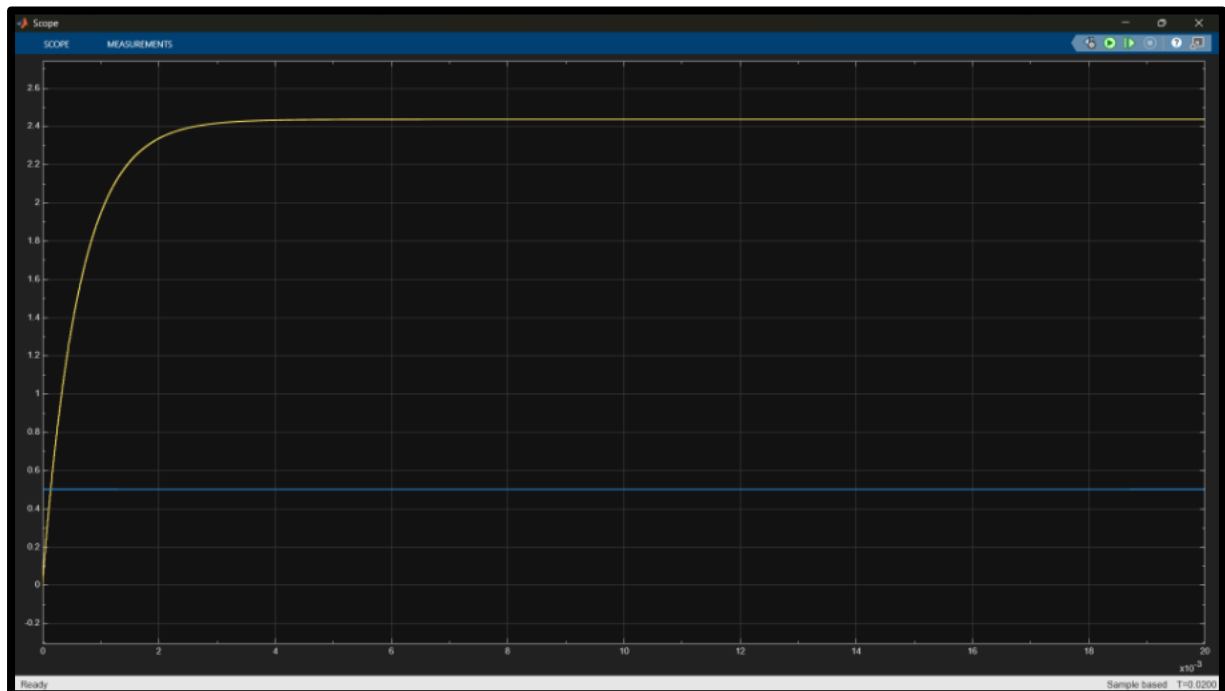
DICKSON VOLTAGE MULTIPLIER

Then we got to know about Dickson circuit then we decided let's try it and will compare the results of both the circuits.

So, its circuit is as follows:



And its output voltage waveform:



So, Dickson circuit in this case we gave **DC supply** instead of an AC source and uses two out-of-phase clock signals (square waves) to push charge along the chain.

A RESEARCH PAPER ON THIS VOLTAGE MULTIPLIER CIRCUITS

Then we got a research paper from the net about these two voltage multipliers Cockcroft and Dickson and comparison as well as to overcome the disadvantages of both multipliers and those researchers made their own voltage multiplier which is the hybrid version of the Dickson and Cockcroft voltage multipliers.

Which they say can overcome the disadvantage of previous both multipliers.

So, we decided first to simulate Full-Wave Cockcroft-Walton Voltage Multiplier and Dickson voltage multiplier and then their hybrid voltage multiplier.

And test it whether the results are good.

Disadvantages:

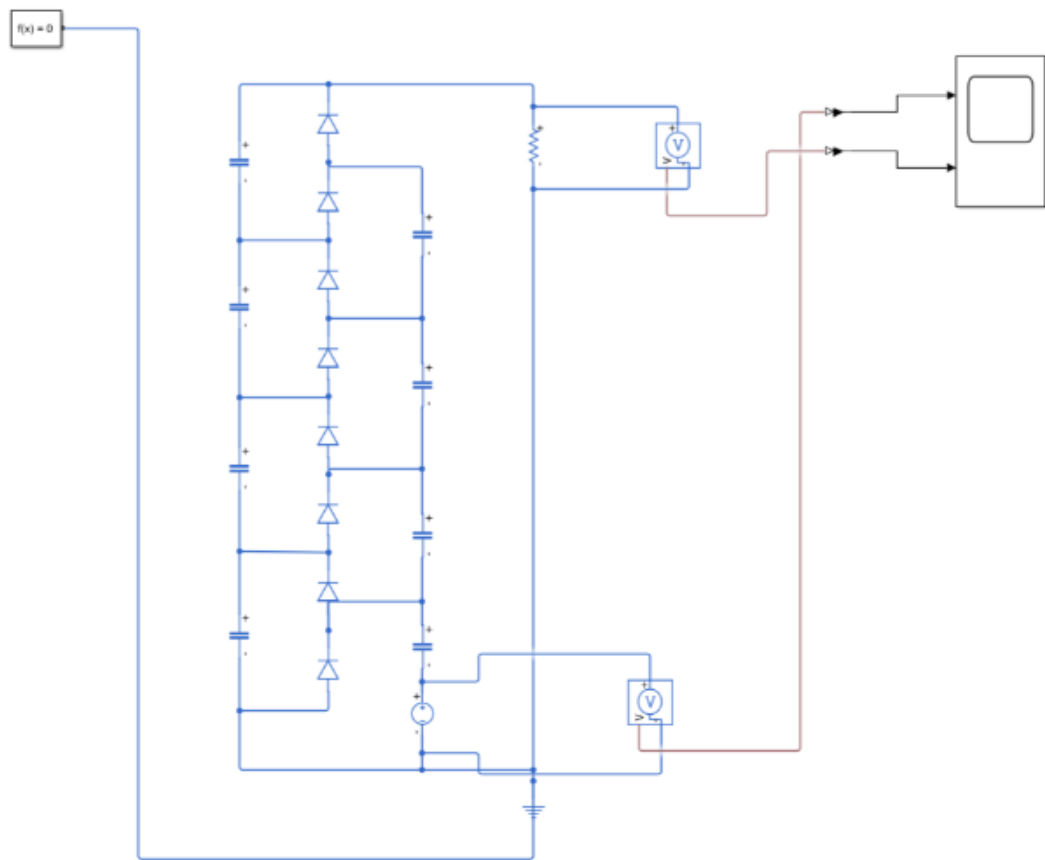
1] Cockcroft-Walton Voltage Multiplier:

- a) High Output Voltage Drop**
- b) Low power efficiency**
- c) High Output Voltage Ripple**
- d) Low Stored Energy**

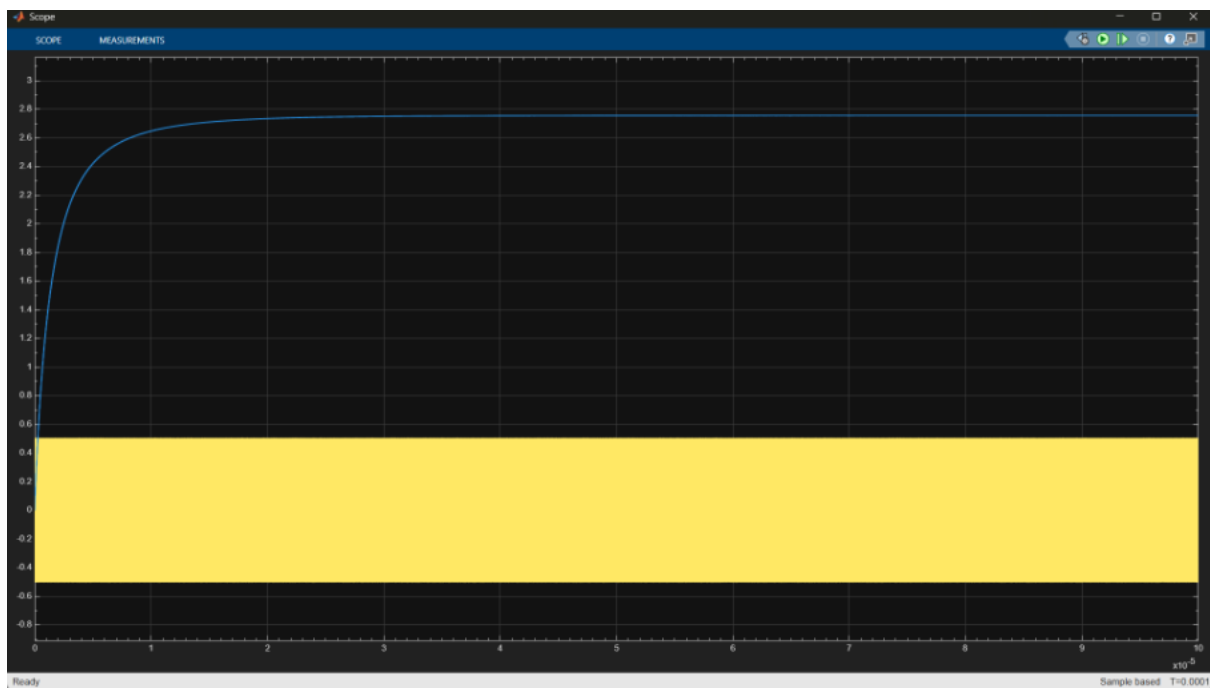
2] Dickson Voltage Multiplier:

- a) High Capacitor Voltage Stress**

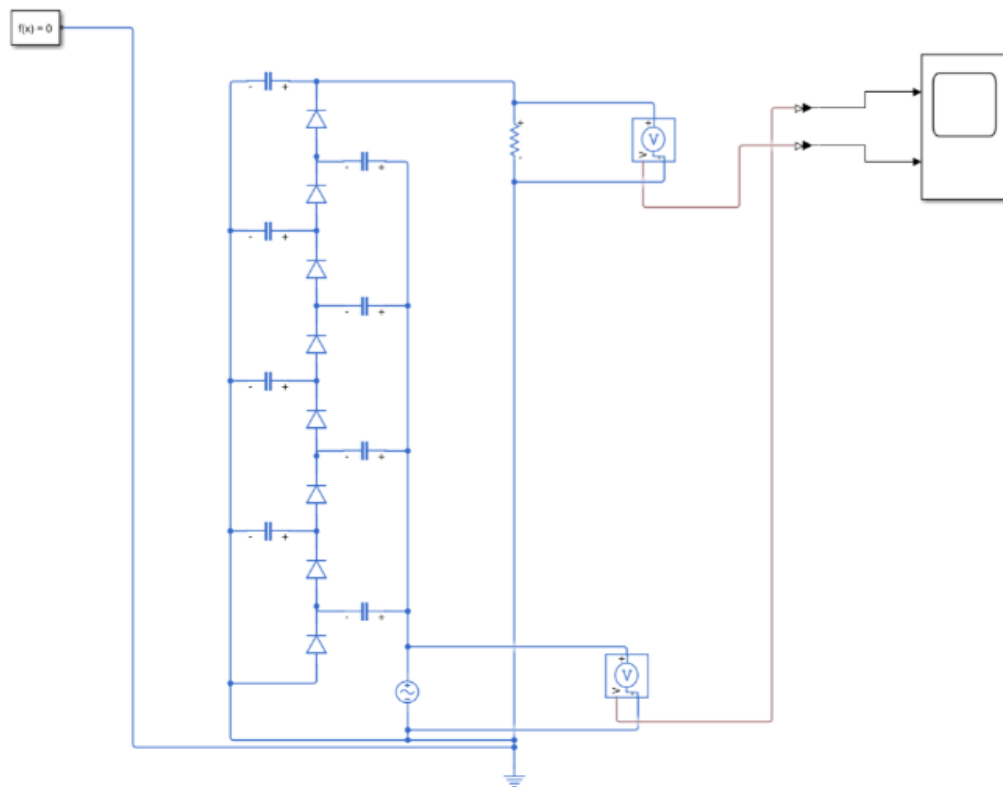
Full Wave Cockcroft-Walton Voltage Multiplier from the Research Paper:



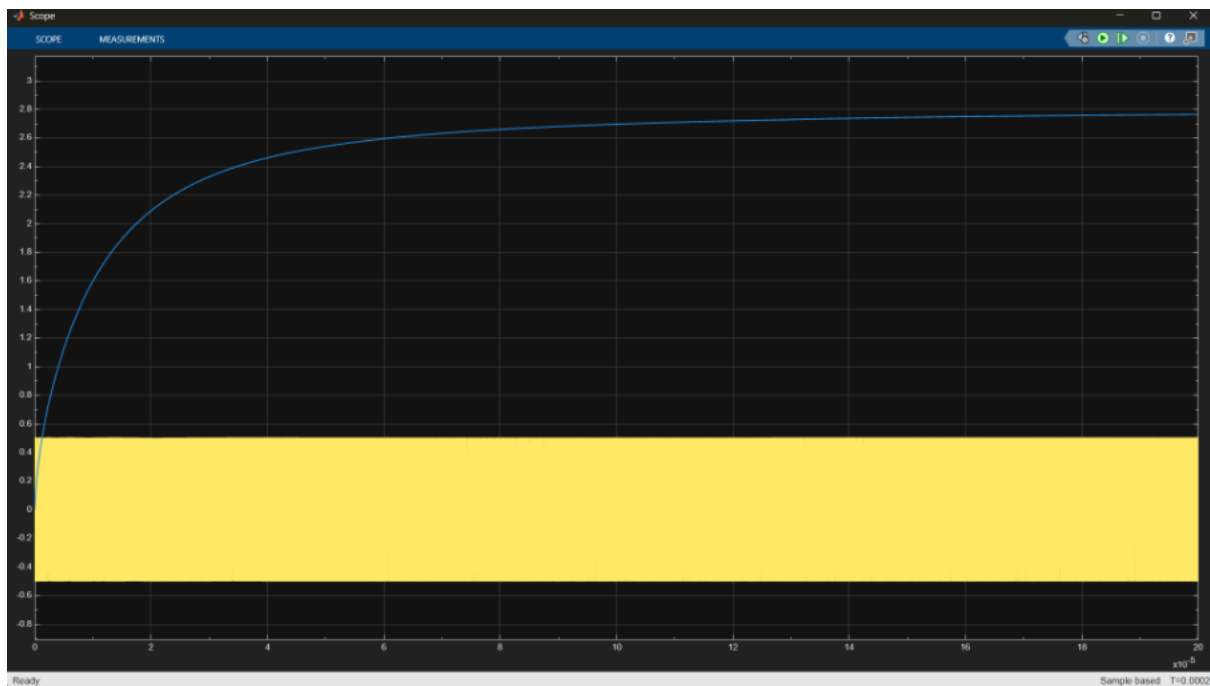
Output waveform:



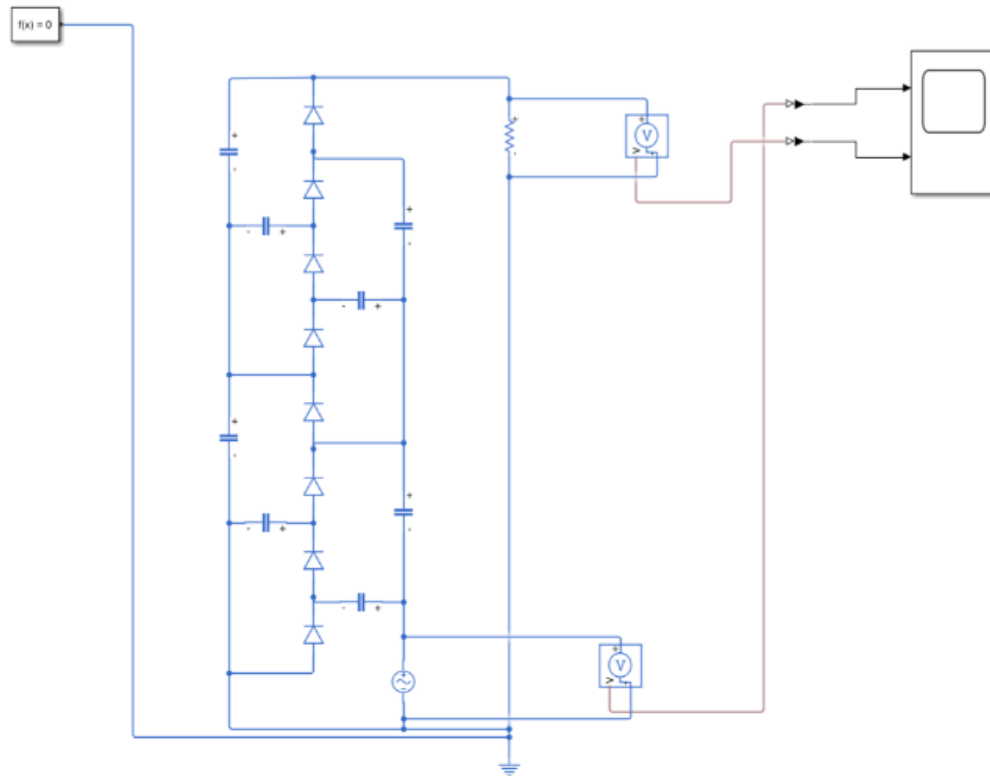
Full Wave Dickson Voltage Multiplier from the Research paper:



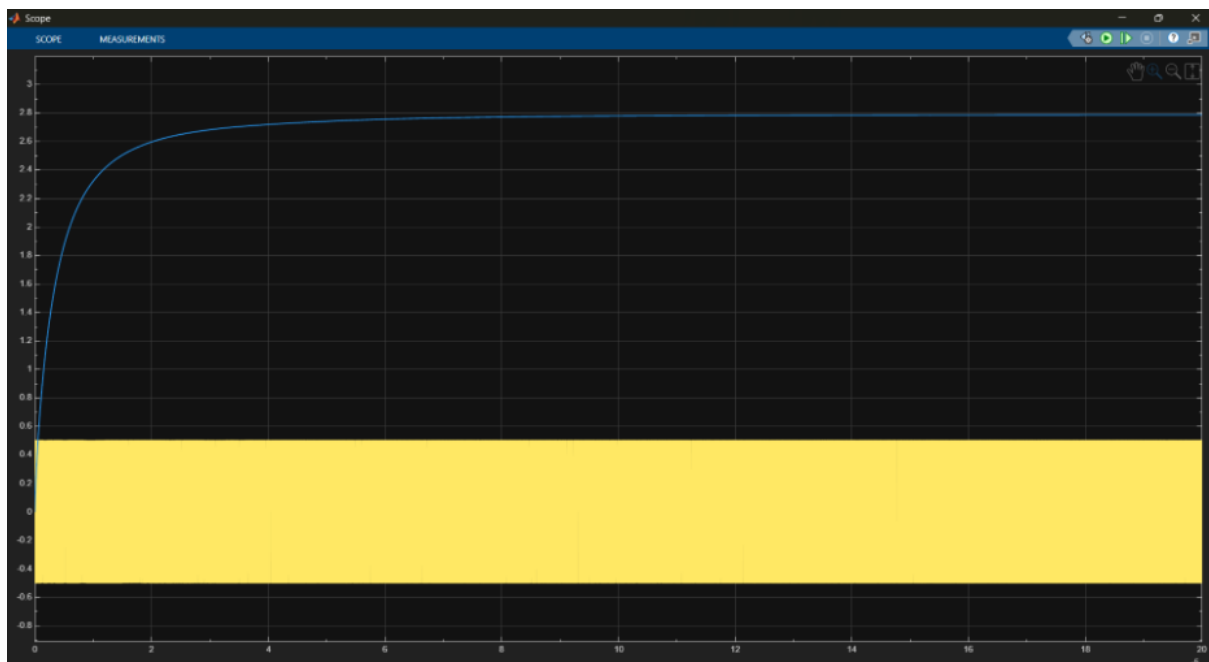
Output Waveform:



Full Wave Hybrid Voltage Multiplier from the Research paper:



Output Waveform:



THE COMPARISON ACCORDING TO RESEARCH PAPER AND SIMULATIONS:

Feature	Cockcroft-Walton	Dickson	Proposed Hybrid
Capacitor Arrangement	All connected in series	All connected in parallel	Combination of series and parallel
Output Voltage Drop	Highest (Significant sag)	Lowest	Intermediate
Capacitor Voltage Stress	Lowest (Uniform across capacitors)	Highest (At the last stage)	Intermediate (Reduced by m times vs. Dickson)
Output Voltage Ripple	Highest	Lowest	Intermediate
Power Efficiency	Lowest	Highest	Intermediate
Stored Energy (E_{cap})	Lowest	Highest	Intermediate

CONCLUSION:

So, after all these research and performing Matlab-Simulations of all those Cockcroft-Walton , Dickson and Hybrid Voltage Multiplier's circuits we have decided to test both Dickson and Hybrid version of voltage multiplier circuits on real hardware with components and see the result and choose the best one for our Project AURA.